Daily Problems in Quantum Information

A worked collection

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Abstract

A running collection of short problems in quantum information, matrix analysis and mathematical physics, each intended to be solvable in roughly ten to fifteen minutes once the central idea is found, together with worked solutions. Each entry records its origin (sourced, extracted, adapted, or original) and, where applicable, a precise reference. Solutions are my own unless stated otherwise; errors are mine.

A *day* here means a day on which I worked, not a calendar day since starting: the numbering is contiguous, the dates are not. Every problem carries a permanent identifier `Qnnn` which never changes, so it can be cited from anywhere — another entry, the toolbox, or the website — independently of where it sits in this book.

Contents

The record iv

Day 1 — 4 September 2026 1
1.1 Pure fixed points of a channel (Q001) 1

Day 2 — 5 September 2026 2
2.1 Squaring is not operator monotone (Q002) 2

Day 3 — 9 September 2026 3
3.1 What survives a channel, and what doesn't (Q003) 3

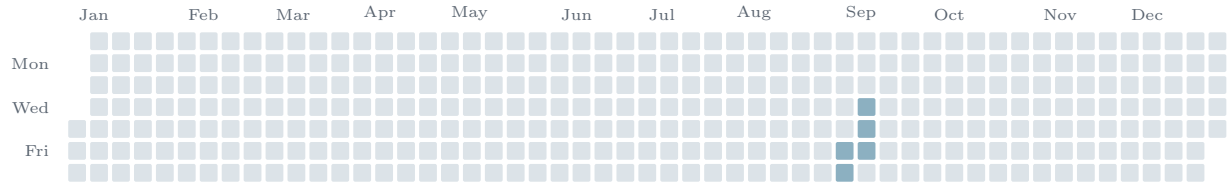
Day 4 — 10 September 2026 4
4.1 A pure marginal forces a product (Q004) 4

Day 5 — 11 September 2026 5
5.1 Two POVMs that never disagree (Q005) 5
5.2 No chain of singlets (Q006) 5

Day 6 — 12 September 2026 6
6.1 Day 7 title (Q007) 6

A Toolbox 7

The record



7 problems over 5 working days.

ID	Day	Date	Title	Topic	Time	D
Q001	1	4 September 2026	Pure fixed points of a channel	Quantum channels; Kraus operators; positive semidefinite arguments	45 min	3
Q002	2	5 September 2026	Squaring is not operator monotone	Operator inequalities; matrix analysis; positivity vs. trace positivity	35 min	3
Q003	3	9 September 2026	What survives a channel, and what doesn't	Trace norm vs. Hilbert–Schmidt norm; positive maps; distinguishability	∼1 h	2
Q004	4	10 September 2026	A pure marginal forces a product	Bipartite states; supports; positivity	20 min	2
Q005	5	11 September 2026	Two POVMs that never disagree	Measurements; POVMs; operator positivity	20 min	1
Q006	5	–	No chain of singlets	Many-body spin chains; monogamy of entanglement; reduced states	–	–
Q007	6	–	Day 7 title		–	–

Day 1

Friday, 4 September 2026

1.1 Pure fixed points of a channel

ID	`Q001`
Topic	Quantum channels; Kraus operators; positive semidefinite arguments
Estimated time	∼12 minutes
Origin	Original
Source	Original problem
Prerequisites	Kraus representation of a channel
Status	written-up

Problem. [**TODO:** paste the problem statement from days/day01.tex]

Solution. [**TODO:** paste the solution from days/day01.tex]

Takeaway. [**TODO:** one or two sentences]

Attempt log

Date	4 September 2026
Time	45 min
Difficulty	3 / 5

Day 2

Saturday, 5 September 2026

2.1 Squaring is not operator monotone

ID	`Q002`
Topic	Operator inequalities; matrix analysis; positivity vs. trace positivity
Estimated time	∼12 minutes
Origin	Adapted
Source	R. Bhatia, *Matrix Analysis*, Springer GTM 169 (1997), Ch. V
Prerequisites	None beyond $M \geq 0$ meaning positive semidefinite
Status	written-up

Problem. [**TODO:** paste the problem statement from days/day02.tex]

Solution. [**TODO:** paste the solution from days/day02.tex]

Takeaway. [**TODO:** one or two sentences]

Attempt log

Date	5 September 2026
Time	35 min
Difficulty	3 / 5

Day 3

Wednesday, 9 September 2026

3.1 What survives a channel, and what doesn't

ID	`Q003`
Topic	Trace norm vs. Hilbert–Schmidt norm; positive maps; distinguishability
Estimated time	∼12 minutes
Origin	Adapted
Source	J. Watrous, *The Theory of Quantum Information* (CUP, 2018), Cor. 3.40
Status	written-up

Problem. [**TODO:** paste the problem statement from days/day03.tex]

Solution. [**TODO:** paste the solution from days/day03.tex]

Takeaway. [**TODO:** one or two sentences]

Attempt log

Date	9 September 2026
Time	∼1 h
Difficulty	2 / 5

Day 4

Thursday, 10 September 2026

4.1 A pure marginal forces a product

ID	`Q004`
Topic	Bipartite states; supports; positivity
Estimated time	∼10 minutes
Origin	Original
Source	Original problem
Prerequisites	Partial trace
Status	written-up

Problem. [**TODO:** paste the problem statement from days/day04.tex]

Solution. [**TODO:** paste the solution from days/day04.tex]

Takeaway. [**TODO:** one or two sentences]

Attempt log

Date	10 September 2026
Time	20 min
Difficulty	2 / 5

Day 5

Friday, 11 September 2026

5.1 Two POVMs that never disagree

ID Q005
Topic Measurements; POVMs; operator positivity
Estimated time ∼10 minutes
Origin Original
Source Original problem
Prerequisites A POVM is a family $\{E_a\}$ with $E_a \geq 0$ and $\sum_a E_a = \mathbb{1}$
Status written-up

Problem. [**TODO:** paste the problem statement from days/day05.tex]

Solution. [**TODO:** paste the solution from days/day05.tex]

Takeaway. [**TODO:** one or two sentences]

Attempt log

Date 11 September 2026
Time 20 min
Difficulty 1 / 5

5.2 No chain of singlets

ID Q006
Topic Many-body spin chains; monogamy of entanglement; reduced states
Estimated time ∼10 minutes
Origin Original
Source Original problem
Prerequisites The singlet $|\Psi^-\rangle$; the Q004 (Day 4) result may be used without reproving it
Status written-up

Problem. [**TODO:** paste the problem statement from days/day06.tex]

Solution. [**TODO:** paste the solution from days/day06.tex]

Takeaway. [**TODO:** one or two sentences]

Day 6

Saturday, 12 September 2026

6.1 Day 7 title

ID	`Q007`
Estimated time	∼12 minutes
Origin	Original
Source	Original problem
Status	written-up

Problem. [**TODO:** paste the problem statement from days/day07.tex]

Solution. [**TODO:** paste the solution from days/day07.tex]

Takeaway. [**TODO:** one or two sentences]

A Toolbox

Results that have earned a name by recurring. Cite them with `\usetool{T1}`; the back-references below are collected automatically.

T1. Vanishing expectation of a positive operator. *If $A \geq 0$ and $\langle\psi|A|\psi\rangle = 0$, then $A|\psi\rangle = 0$, and hence also $\langle\psi|A = 0$.*

Used in: not yet

T2. Purity of a marginal forces a product. *If $\rho \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ has a pure marginal $\rho_A = |\psi_A\rangle\langle\psi_A|$, then $\rho = |\psi_A\rangle\langle\psi_A| \otimes \rho_B$.*

Used in: not yet